push_i
pred_in_i[0]
accept_o
&
push_w
有效位
控制
pop1_w
pop2_w
accept_o
count_q ≠ 2
count_q
data_in_i[63:0]
pc_in_i
info0/1_in_i
csr0/1_in_i
push_w
rst_i / flush_i
accept_o
push_w:
1 / ~pred_in_i[0]
取指包存储（DEPTH = 2）

	ram_q 64 bit	pc_q 32 bit	info0_q / info1_q	csr_history_q	valid0_q / valid1_q
0	ram_q[0]	pc_q[0]	info0_q[0] info1_q[0]	csr_history_q[0]	valid0_q[0] valid1_q[0]
1	ram_q[1]	pc_q[1]	info0_q[1] info1_q[1]	csr_history_q[1]	valid0_q[1] valid1_q[1]

wr_ptr_q
push_w
队列控制
wr_ptr_q
rd_ptr_q
count_q
count_q ≠ 0
rd_ptr_q
读出选择
有效判定
valid0_o
&
pop0_i
valid1_o
&
pop1_i
valid0_o / valid1_o
pop1_w
pop2_w
整包消费
完成判定
pop_complete_w
data0_out_o[31:0]
pc0_out_o
info0_out_o / csr0_out_o
data1_out_o[31:0]
pc1_out_o
info1_out_o / csr1_out_o
pc0_out_o = {pc_q[rd_ptr_q][31:3], 3'b000}
pc1_out_o = {pc_q[rd_ptr_q][31:3], 3'b100}